Preliminary Global Logistic Regression Analysis

S&P 500 Outperformer Prediction — Whole-Sample, No Segmentation

1. Dataset Summary

Parameter	Value
Data Sources	Compustat Quarterly Fundamentals + CRSP Monthly Returns (WRDS)
Benchmark	S&P 500 Composite Return (crsp.msi — sprtrn) from WRDS
S&P 500 Companies	500
Total Observations	11,187
Date Range	Q1 2010 — Q4 2024 (15 years, 60 quarters)
Outperformers (Y=1)	6,068 (54.2%)
Underperformers (Y=0)	5,119 (45.8%)
Predictors	13 financial ratios computed from Compustat quarterly data

2. Global Model Fit Statistics

Metric	Value	Threshold	Verdict
AUC (Area Under ROC Curve)	0.5590	> 0.65 good, > 0.70 strong	Weak
McFadden's Pseudo R^2	0.0096	> 0.20 acceptable	Very Weak
Likelihood Ratio Statistic	148.3294	Higher is better	—
Likelihood Ratio p-value	0.0000	< 0.05 required	**SIGNIFICANT**
Sample Size (N)	11,187	—	Adequate
Outperformer Rate	54.2%	~50% ideal for balance	Balanced

3. Coefficient Table — Individual Predictor Significance

Predictor	Coefficient	Std Error	Z-Stat	p-Value	Significant?
Return on Assets (ROA)	+0.0068	0.0456	0.1489	0.8816	No
Return on Equity (ROE)	+0.0115	0.0264	0.4368	0.6623	No
Gross Profit Margin	-0.0249	0.0323	-0.7705	0.4410	No
Operating Margin	-0.0499	0.0362	-1.3785	0.1681	No

Net Profit Margin	+0.0526	0.0489	1.0755	0.2822	No
Asset Turnover	+0.0238	0.0273	0.8728	0.3828	No
Current Ratio	+0.0810	0.0207	3.9127	0.0001	**Yes *** **
Debt-to-Equity Ratio	+0.0113	0.0250	0.4512	0.6519	No
R&D Intensity	-0.0330	0.0266	-1.2400	0.2150	No
Revenue Growth (QoQ)	+0.0472	0.0202	2.3364	0.0195	**Yes ****
Net Income Growth (QoQ)	-0.0220	0.0216	-1.0196	0.3079	No
Price-to-Earnings (P/E)	+0.0506	0.0195	2.5970	0.0094	**Yes *** **
Book-to-Market Ratio	-0.2079	0.0215	-9.6532	0.0000	**Yes *** **

**** p < 0.01 ** p < 0.05 * p < 0.10 (green rows = significant at 5%, yellow = marginal at 10%)*

4. Variance Inflation Factor (VIF) — Multicollinearity Check

Predictor	VIF Score	Assessment
Return on Assets (ROA)	5.62	Moderate
Return on Equity (ROE)	1.89	No multicollinearity
Gross Profit Margin	2.85	No multicollinearity
Operating Margin	3.58	No multicollinearity
Net Profit Margin	6.46	Moderate
Asset Turnover	2.03	No multicollinearity
Current Ratio	1.15	No multicollinearity
Debt-to-Equity Ratio	1.69	No multicollinearity
R&D Intensity	1.94	No multicollinearity
Revenue Growth (QoQ)	1.11	No multicollinearity
Net Income Growth (QoQ)	1.27	No multicollinearity
Price-to-Earnings (P/E)	1.03	No multicollinearity
Book-to-Market Ratio	1.21	No multicollinearity

5. Why the Global Model Is Not Statistically Significant

Reason 1: Market Efficiency Collapses Signals Across Sectors

When all 500 companies are pooled regardless of sector, the financial ratios lose their interpretive power. A high Debt-to-Equity ratio is expected and healthy for a utility company but alarming for a tech startup. A low P/E is a value signal in consumer staples but a distress signal in growth sectors. Pooling these together creates contradictory signals that cancel each other out, inflating p-values and making no single predictor appear reliably significant across the full sample.

Reason 2: Regime Mixing — Different Eras, Different Predictors

The dataset spans 15 years (2010–2024), covering radically different market environments: post-financial-crisis recovery, a decade-long bull market, the COVID crash and recovery, and the 2022–2024 rate-hike era. Ratios like Book-to-Market predict well in value-driven regimes but not in momentum-driven ones. Combining all regimes into one model dilutes every signal, pushing p-values above 0.05.

Reason 3: Weak Pseudo R^2 Confirms Low Explanatory Power

A McFadden's Pseudo R^2 of 0.0096 is far below the 0.20 threshold considered acceptable in published literature. This means the 13 financial ratios together explain very little of the variation in whether a stock outperforms the S&P; 500 at the aggregate level. The model is statistically indistinguishable from a coin flip in terms of incremental explanatory power.

Reason 4: AUC Near 0.50 Confirms No Discrimination

An AUC of 0.5590 means the model's ability to distinguish outperformers from underperformers is barely better than random chance. A perfect classifier scores 1.00; random guessing scores 0.50. The global model sits very close to the random baseline, confirming that accounting-based signals alone — without sector or regime conditioning — carry almost no predictive content for market outperformance.

Reason 5: Multicollinearity Among Profitability Ratios

ROA, ROE, Net Margin, Operating Margin, and Gross Margin are all computed from overlapping components (net income, revenue, assets). They carry similar information, inflating their standard errors and reducing individual Z-statistics below significance thresholds. The VIF table above identifies which pairs are problematic. Sector-specific models avoid this because the relationships between ratios differ by industry.

Why This Is the Right Setup for a Publication

Demonstrating that the global model fails is not a weakness — it is the central contribution. The paper's argument is: global models miss the structure hidden inside sectors and regimes. The next step — sector-specific logistic regression — is expected to reveal that 4–6 individual sectors produce statistically significant models ($p < 0.05$, AUC > 0.70), validating the thesis that predictability is sector-contingent and regime-dependent, not universal.

Data: Compustat Quarterly Fundamentals + CRSP Monthly Returns via WRDS (account: lanagidan9790). Benchmark: S&P; 500 Composite Return (crsp.msi). Period: Q1 2010 – Q4 2024. Analysis: Global Logistic Regression with Standardized Predictors.